

E-commerce Analytics Report

Generated May 28, 2026 | AI Analytics Platform

Key Performance Indicators

Total Records 5,000	Avg Order Value \$538.4K
Avg Age 48.16	Avg Total Orders 10.03
Avg Last Purchase 180.07	Avg Is Fraudulent 0.03
Avg Email Open Rate 50.73	Avg Loyalty Score 50.04
Avg Churn Risk 0.28	Date Range Jun 2020 – Jun 2025

Executive Summary

This E-commerce Analytics report analyses synthetic_ecommerce_churn_dataset.csv, covering 0 records across 0 dimensions. The dataset has been automatically cleaned, validated, and scored for quality prior to analysis.

Key performance indicators show: Total Records at 5,000, Avg Order Value at \$538.4K, Avg Age at 48.16, Avg Total Orders at 10.03. No strong directional trends were detected in the current dataset. The domain-specific KPI framework applied is aligned to E-commerce Analytics best practices, ensuring that the metrics presented reflect operationally relevant performance indicators rather than generic statistical averages.

Anomaly detection flagged 4 area(s) requiring attention. These may indicate data quality issues, outlier events, or genuine business extremes that warrant further investigation before operational decisions are made. Particular attention should be given to columns flagged during quality scoring, as these may affect the reliability of aggregated KPI figures.

To unlock AI-generated narrative insights, executive commentary, and natural language recommendations, configure your API key: set ANTHROPIC_API_KEY (Anthropic Claude) or OPENAI_API_KEY (OpenAI GPT-4o) in the .env file and restart the application. The AI layer will then provide domain-specific strategic analysis, anomaly explanations, and prioritised action plans tailored to your E-commerce Analytics context.

Visualisations

Avg Order Value Over Time

Chart export failed for 'Avg Order Value Over Time': expected str, bytes or os.PathLike object, not ImageReader

Last Purchase Over Time

Chart export failed for 'Last Purchase Over Time': expected str, bytes or os.PathLike object, not ImageReader

Avg Order Value by Gender

Chart export failed for 'Avg Order Value by Gender': expected str, bytes or os.PathLike object, not ImageReader

Avg Order Value by Country

Chart export failed for 'Avg Order Value by Country': expected str, bytes or os.PathLike object, not ImageReader

Distribution: Avg Order Value

Chart export failed for 'Distribution: Avg Order Value': expected str, bytes or os.PathLike object, not ImageReader

Avg Order Value vs Last Purchase

Chart export failed for 'Avg Order Value vs Last Purchase': expected str, bytes or os.PathLike object, not ImageReader